Example 2.22 The equations $x^2 - 6x + a = 0$ and $x^2 - bx + 6 = 0$ have one root in common. The other root of the first and the second equations are integers in the ratio $4 : 3$. Find the common root.

Solution:
Let α be the common root.
Let $\alpha, 4\beta$ be the roots of $x^2 - 6x + a = 0$.
Let $\alpha, 3\beta$ be the roots of $x^2 - bx + 6 = 0$.
Then, $4\alpha\beta = a$ and $3\alpha\beta = 6$ which give $\alpha\beta = 2$ and $a = 8$.
The roots of $x^2 - 6x + 8 = 0$ are $2, 4$.
If $\alpha = 2$, then $\beta = 1$
If $\alpha = 4$, then $\beta = \frac{1}{2}$ which is not an integer.
Hence, the common root is 2.

Example 2.23 Find the values of p for which the difference between the roots of the equation $x^2 + px + 8 = 0$ is 2.

Solution:
Let α and β be the roots of the equation $x^2 + px + 8 = 0$.
Then, $\alpha + \beta = -p$, $\alpha\beta = 8$ and $|\alpha - \beta| = 2$.
Now, $(\alpha + \beta)^2 - 4\alpha\beta = (\alpha - \beta)^2$, which gives $p^2 - 32 = 4$. Thus, $p = \pm 6$.

Exercise - 2.7

1. Factorize: $x^4 + 1$. (*Hint:* Try completing the square.)
2. If $x^2 + x + 1$ is a factor of the polynomial $3x^3 + 8x^2 + 8x + a$, then find the value of a.

2.7 Rational Functions

A rational expression of x is defined as the ratio of two polynomials in x, say $P(x)$ and $Q(x)$ where $Q(x) \neq 0$. Examples of rational expressions are $\dfrac{2x+1}{x^2+x+1}$, $\dfrac{x^4+1}{x^2+1}$ and $\dfrac{x^2+x}{x^2-5x+6}$.

If the degree of the numerator $P(x)$ is equal to or larger than that of the denominator $Q(x)$, then we can write $P(x) = f(x)Q(x) + r(x)$ where $r(x) = 0$ or the degree of $r(x)$ is less than that of $Q(x)$.
So $\dfrac{P(x)}{Q(x)} = f(x) + \dfrac{r(x)}{Q(x)}$.

2.7.1 Rational Inequalities

Example 2.24 Solve $\dfrac{x+1}{x+3} < 3$.

Solution:
Subtracting 3 from both sides we get $\dfrac{x+1}{x+3} - 3 < 0$.